Anlea Maria Jose, 02, Experiment 7

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```
import numpy as np
from scipy import stats

data = np.array([45, 50, 47, 52, 49, 51, 48, 46, 53, 50, 49,
                 47, 52, 51, 48, 46, 50, 49, 47, 53, 52, 48,
                 46, 51, 50, 49, 47, 52, 48, 50])

n = len(data)
x_bar = np.mean(data)
s2 = np.var(data, ddof = 1)
s = np.sqrt(s2)

confidence = 0.95
alpha = 1 - confidence

z_critical = stats.norm.ppf(1-alpha/2)
margin_error = z_critical*s/np.sqrt(n)
mean_lower = x_bar - margin_error
mean_upper = x_bar + margin_error


df = n-1
chi2_lower = stats.chi2.ppf(alpha/2, df)
chi2_upper = stats.chi2.ppf(1-alpha/2, df)
var_lower =(df*s2)/chi2_upper
var_upper = (df*s2)/chi2_lower

print("Sample Data: ", data)
print("Sample Size: ", n)
print("Sample Mean: ", round(x_bar, 4))
print("Sample Variance: ", round(s2, 4))

print("\n95% Confidence Interval for Population Mean: ")
print(f"({round(mean_lower, 4)}, {round(mean_upper, 4)})")

print("\n95% Confidence Interval for Population Variance: ")
print(f"({round(var_lower, 4)}, {round(var_upper, 4)})")
```

```
Sample Data:  [45 50 47 52 49 51 48 46 53 50 49 47 52 51 48 46 50 49 47 53 52 48 46 51
 50 49 47 52 48 50]
Sample Size:  30
Sample Mean:  49.2
Sample Variance:  5.0621

95% Confidence Interval for Population Mean:
(48.3949, 50.0051)

95% Confidence Interval for Population Variance:
(3.2107, 9.1481)
```

```
import numpy as np
from scipy import stats

data = np.array([12, 15, 14, 10, 13, 16, 11, 14, 15, 12])

n = len(data)
x_bar = np.mean(data)
s2 = np.var(data, ddof = 1)
```

```
s = np.sqrt(s2)


confidence = 0.95
alpha = 1 - confidence

df = n-1

t_critical = stats.t.ppf(1-alpha/2, df)
margin_error_mean = t_critical*s/np.sqrt(n)
mean_lower = x_bar - margin_error_mean
mean_upper = x_bar + margin_error_mean

chi2_lower = stats.chi2.ppf(alpha/2, df)
chi2_upper = stats.chi2.ppf(1-alpha/2, df)
var_lower =(df*s2)/chi2_upper
var_upper = (df*s2)/chi2_lower

print("Sample Data: ", data)
print("Sample Size: ", n)
print("Sample Mean: ", round(x_bar, 4))
print("Sample Variance: ", round(s2, 4))

print("\n95% Confidence Interval for Population Mean: ")
print(f"({mean_lower: .4f}, {mean_upper: .4f})")

print("\n95% Confidence Interval for Population Variance: ")
print(f"({var_lower: .4f}, {var_upper: .4f})")
```

```
Sample Data:  [12 15 14 10 13 16 11 14 15 12]
Sample Size:  10
Sample Mean:  13.2
Sample Variance:  3.7333

95% Confidence Interval for Population Mean:
( 11.8178,  14.5822)

95% Confidence Interval for Population Variance:
( 1.7663,  12.4426)
```